

DSA STUDY BLUEPRINT SERIES

91. Binary Search Trees BSTs

C++ BST Search & Insert Actions

Section 09 • C++ Core Data Structures & Algorithms

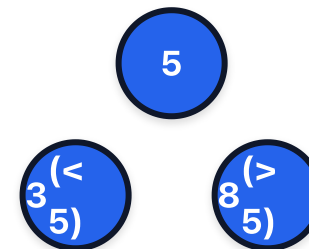
Core Concepts & Visual Blueprint

Theoretical models and memory architectures

Key Principles

- **BST Property:** Left values $<$ Parent $<$ Right values.
- Search divides search space by half recursively.

Memory & Execution Blueprint



C++ Implementation Reference

Standard code layout and syntax

```
TreeNode* insert(TreeNode* root, int val) {  
    if (!root) return new TreeNode(val);  
    if (val < root->val) root->left = insert(root->left, val);  
    else root->right = insert(root->right, val);  
    return root;  
}
```

Algorithmic Complexity & Best Practices

Performance evaluations and critical guidelines

Performance Table

Aspect / Case	Complexity
Search / Insert	$O(\log N)$ average / $O(N)$ skewed

Key Practices & Gotchas

- **Important:** BST operations degrade to $O(N)$ linear time if elements are inserted in sorted order.
- Verify boundary conditions (e.g. empty inputs, single elements).
- Optimize dynamic memory allocations to prevent leaks.

Dry Run & Step-by-Step Execution

Trace analysis on a sample input case

Execution Walkthrough

Let's trace BST constraint validation checks verifying node ranges `[Min, Max]`:

Active Node	Valid range limit	Comparison check	Recursive branch step
15 (Root)	$[-\infty, +\infty]$	True $(-\infty < 15 < +\infty)$	Recurse Left child with range $[-\infty, 15]$
10 (Left)	$[-\infty, 15]$	True $(-\infty < 10 < 15)$	Recurse Right child with range $[10, 15]$

Interview Variations & Optimization Pathways

Common follow-up problems and optimization strategies

Key Variations & Follow-up Questions

- **Inorder property:** BST Inorder traversals yield strictly increasing sorted lists.
- **Tree Balancing:** Balanced tree heights (AVL/Red-black) guarantee logarithmic search times.
- **Related LeetCode Problems:** #98 Validate Binary Search Tree, #700 Search in a Binary Search Tree, #108 Convert Sorted Array to Binary Search Tree.